Innate Immune Cells

Some slides kindly provided by Prof Clare Bennet

Lecture Outline

What is an innate immune cell?

- Innate v. adaptive

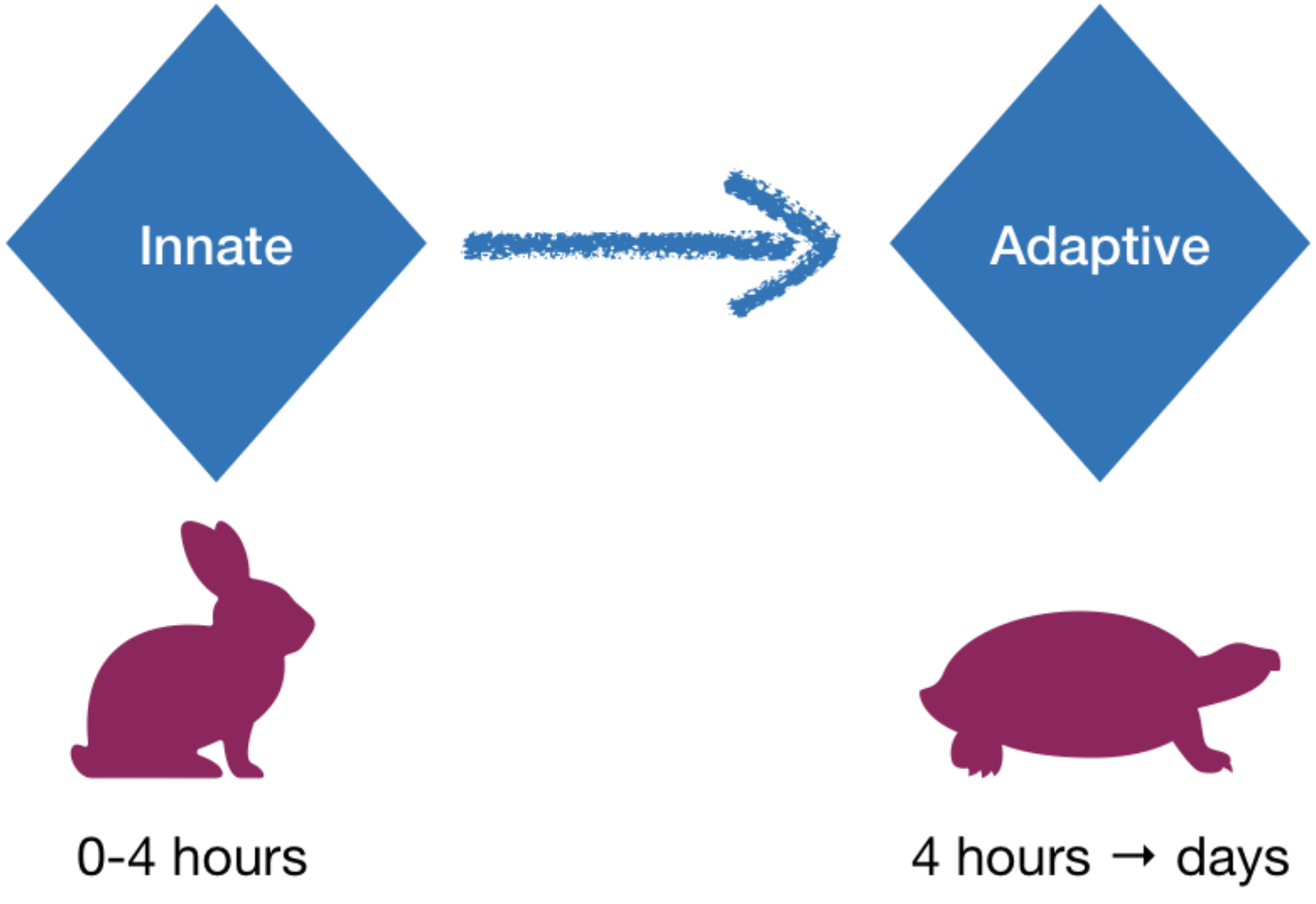
What do innate immune cells do?

- General principles
  - How do immune cells sense danger?
  - How do immune cells communicate with each other?

In detail: myeloid lineage cells

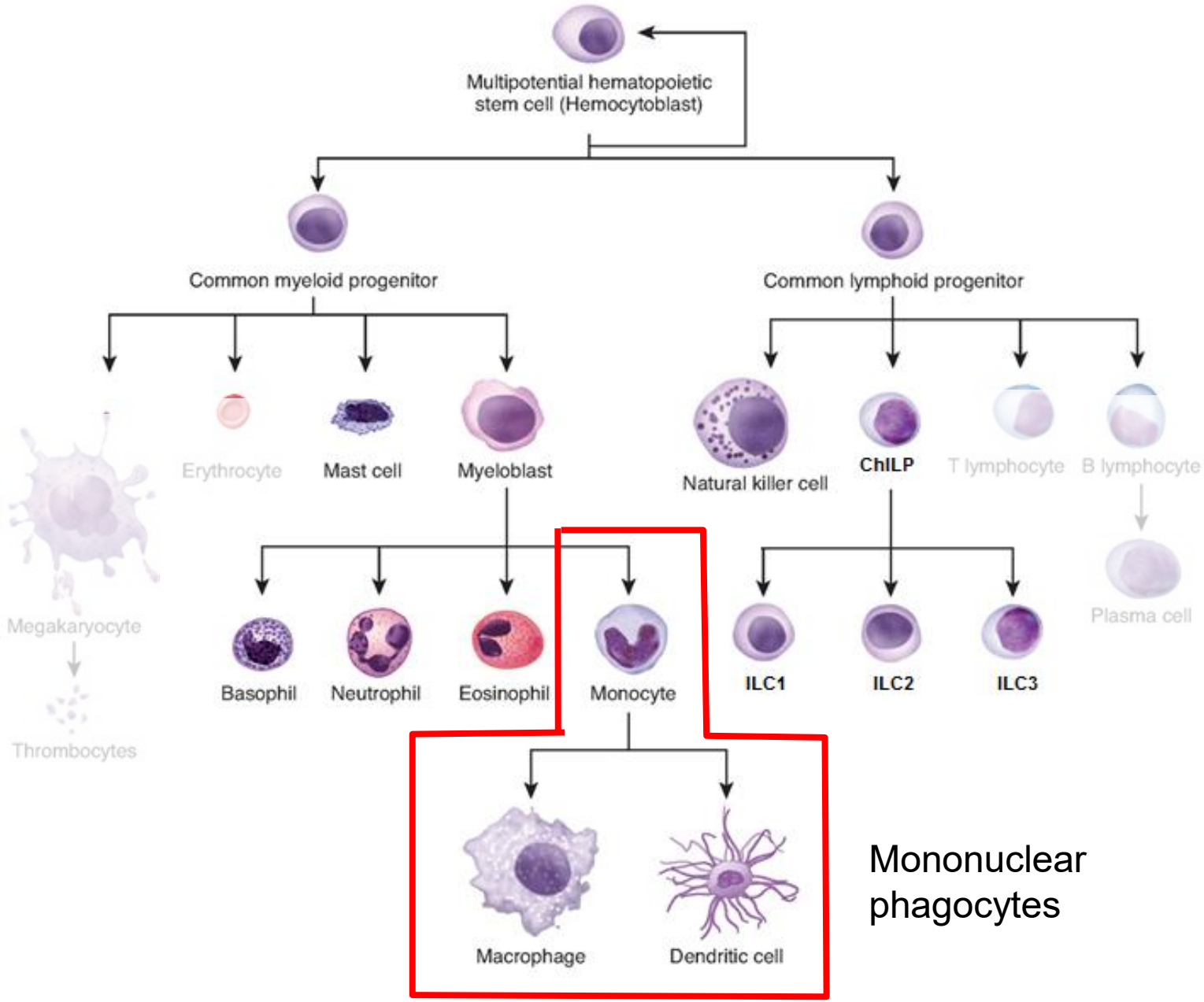
- Neutrophils
- Monocytes
- Macrophages
- Eosinophils and Basophils
- Dendritic cells

What is an innate immune cell?



Memory

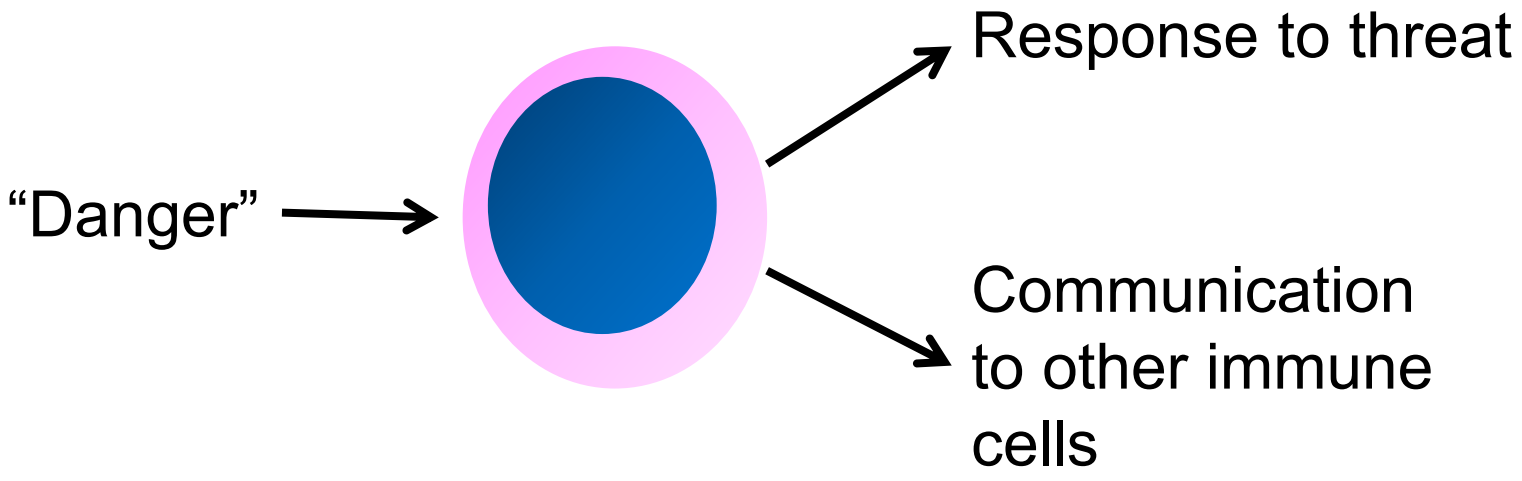
What is an innate immune cell?

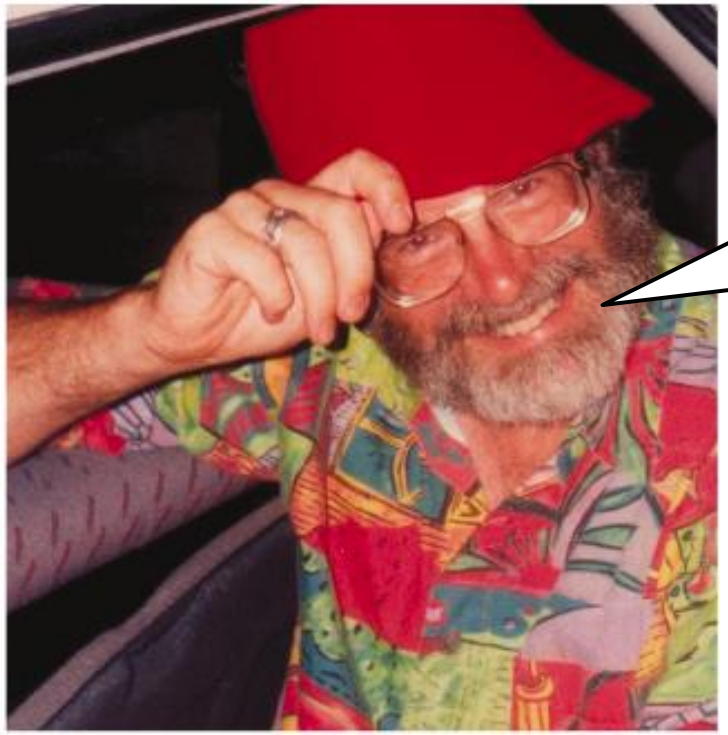


Mononuclear phagocytes

What do innate immune cells do?

Inputs **Outputs**





Charlie Janeway

The immunologist's dirty little secret...

Approaching the Asymptote? Evolution and Revolution in Immunology

C.A. JANEWAY, JR.

Section of Immunology, Howard Hughes Medical Institute at Yale University School of Medicine
New Haven, Connecticut 06510

It is a great privilege to introduce this Cold Spring Harbor Symposium on quantitative biology on the subject of immune recognition. My predecessors in this ...le, Macfarlane Burnet (1967) and Niels Jerne (1976), ...olound effect on the development of ideas in ...ny of the participants have made great ...derstanding of the subject of this ...ore qualified than I to consider ...

most important of these was the role of the MHC in controlling immune responses, and much effort was directed to understanding the MHC molecules and the genes that encode them. Finally, the 1976 symposium saw the first description of immunoglobulin gene rearrangement in antibody-forming cells by Susumu Tonegawa (1976), opening the way to an understanding of the greatest genetic puzzle of the antibody molecule, its possession of variable and constant structural motifs.

APPROACHING THE ASYMPTOTE?

...ear's symposium on immunological recognition ...the tradition of its precursors. Its domi... ...ured in Figure 3 of Madden et al. ...ucture of a class I MHC ...ound to its outer ...ure recog...

How do immune cells know that something is potentially dangerous?

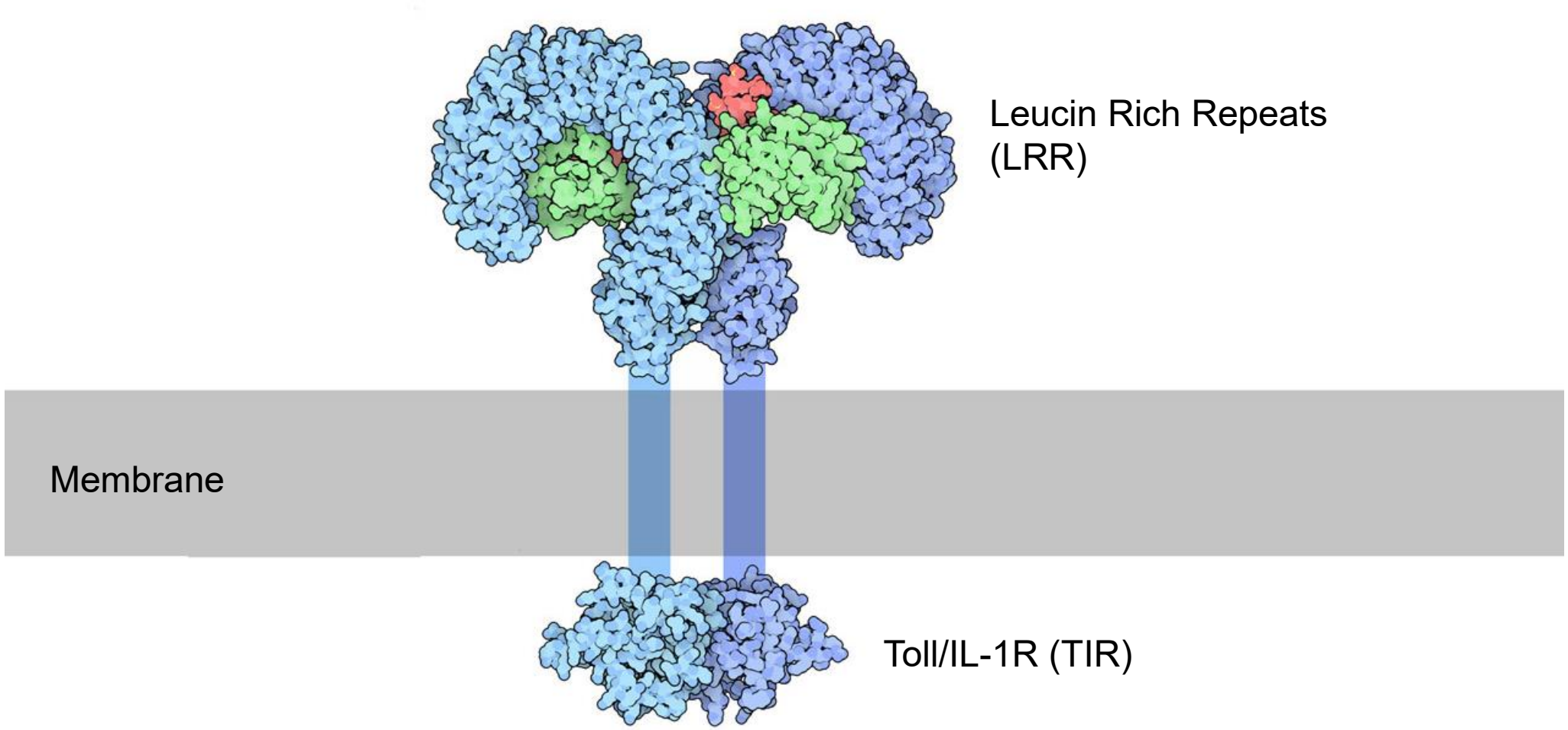
- Pathogen associated molecular patterns (PAMPs)
  - Found in pathogens but not us
  - Highly conserved
  - Essential for pathogen life cycle

- Recognised by pattern recognition receptors (PRRs)
  - Toll-like receptors
  - Nod-like receptors
  - C-type lectin receptors
  - RIG-I and Mda5

Bruce Beutler, 2011

Jules Hoffman, 2011

Toll-like receptors are one kind of PRR...

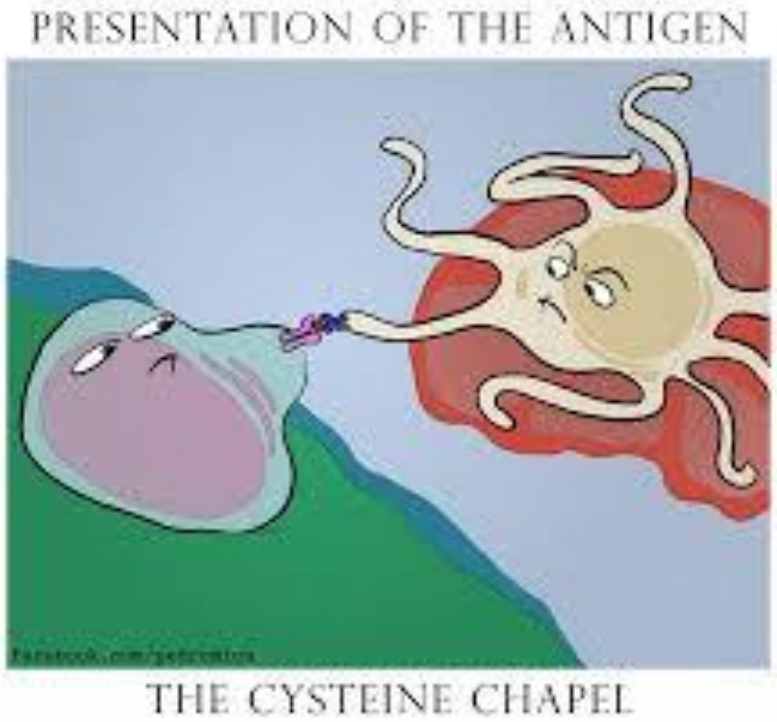


Toll-like receptors are one kind of PRR

	Ligand	Pathogen
TLR1	Lipopeptides	Bacteria
TLR2	Glycolipids, lipopeptides Zymosan	Bacteria Fungi
TLR3	dsRNA	RNA viruses
TLR4	LPS	Gram negative bacteria
TLR5	Flagellin	Bacteria
TLR6	Lipopeptides	Mycoplasma
TLR7	ssRNA	RNA viruses
TLR8	ssRNA	RNA viruses
TLR9	Unmethylated CpG DNA	Bacteria, DNA viruses

TLRs 2 and 4 also recognise host heat shock proteins (from tissue damage)

How do immune cells communicate with each other?



Pedromics (@Pedromics)

- Antigen presentation
- Cytokines

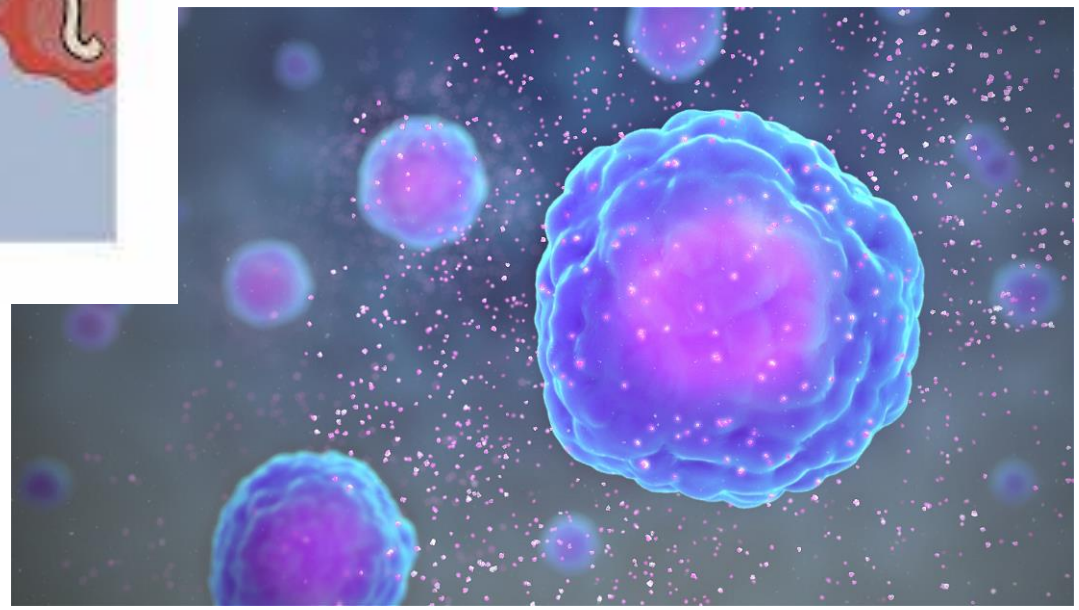


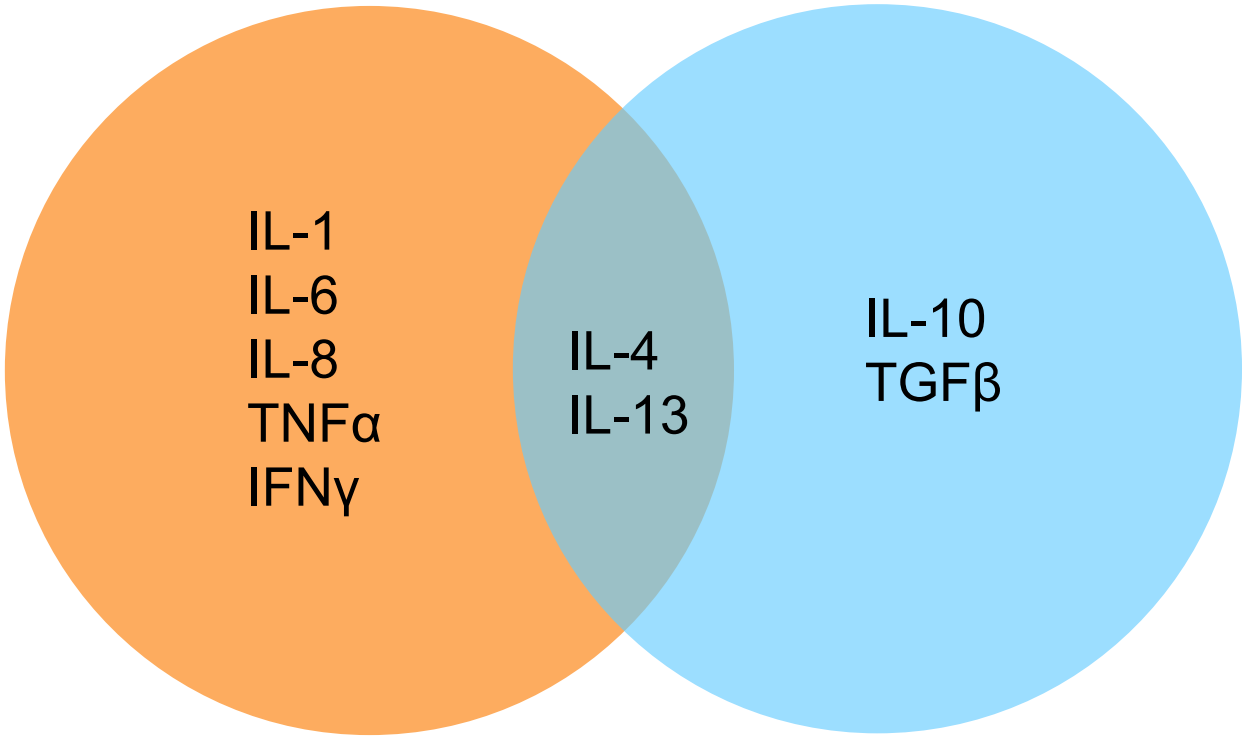
Image from Wikipedia

Cytokines signal between cells during an immune response

- Secreted molecules, either proteins or glycoproteins
- Signal between cells during an immune response
- Only a cell that has the appropriate receptor can respond to a particular cytokine…

Regulation !

Cytokines can be divided into pro-inflammatory and anti-inflammatory



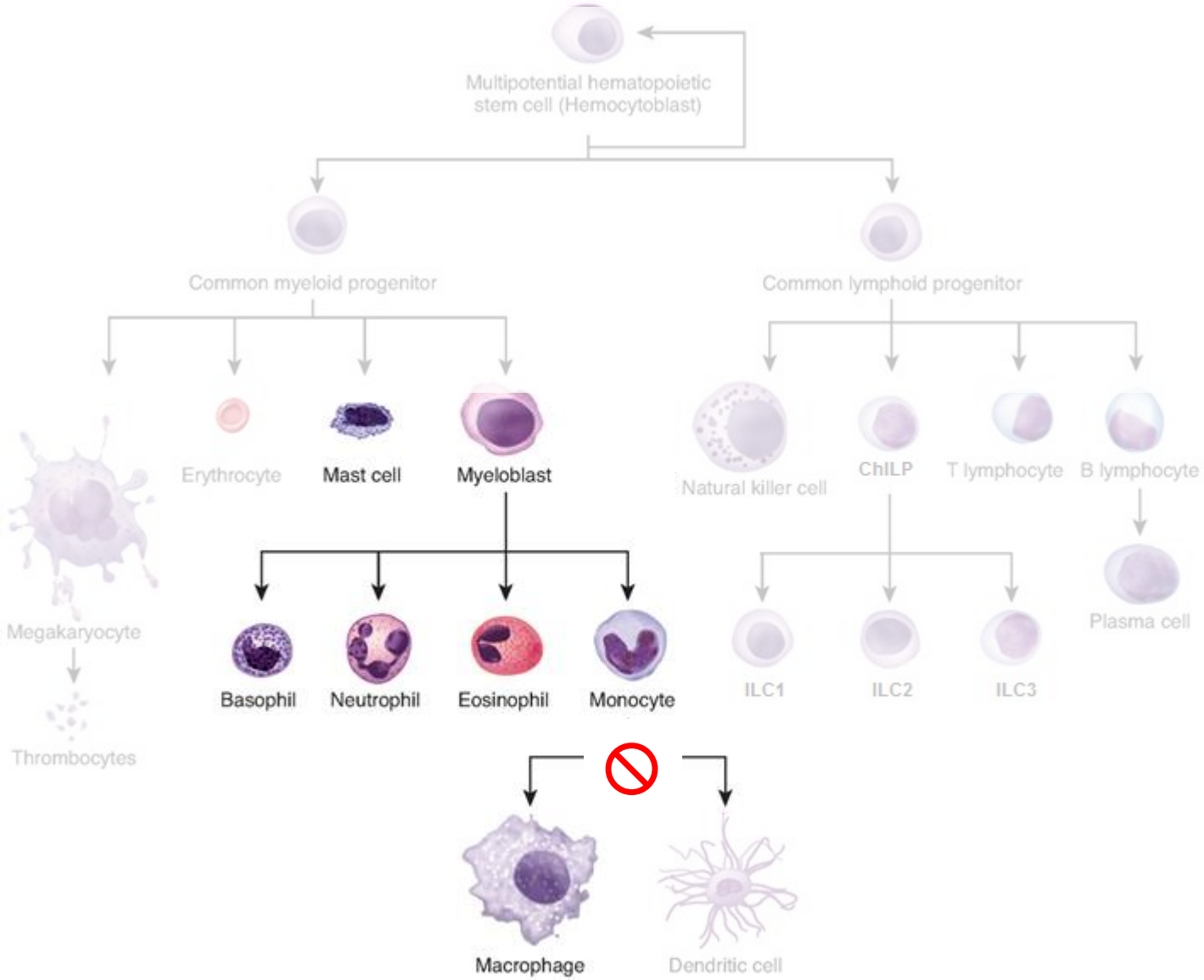
Summary 1 - What do innate immune cells do?

1. Rapidly respond to danger
2. Express receptors that specifically allow them to sense infection or tissue damage
3. Directly respond to threats
4. Produce cytokine signals to recruit other immune cells

Part 2: Myeloid lineage cells

Neutrophils
Monocytes
Macrophages
Basophils & Eosinophils

The myeloid lineage



We can separate innate myeloid cells by flow cytometry based on their size and shape

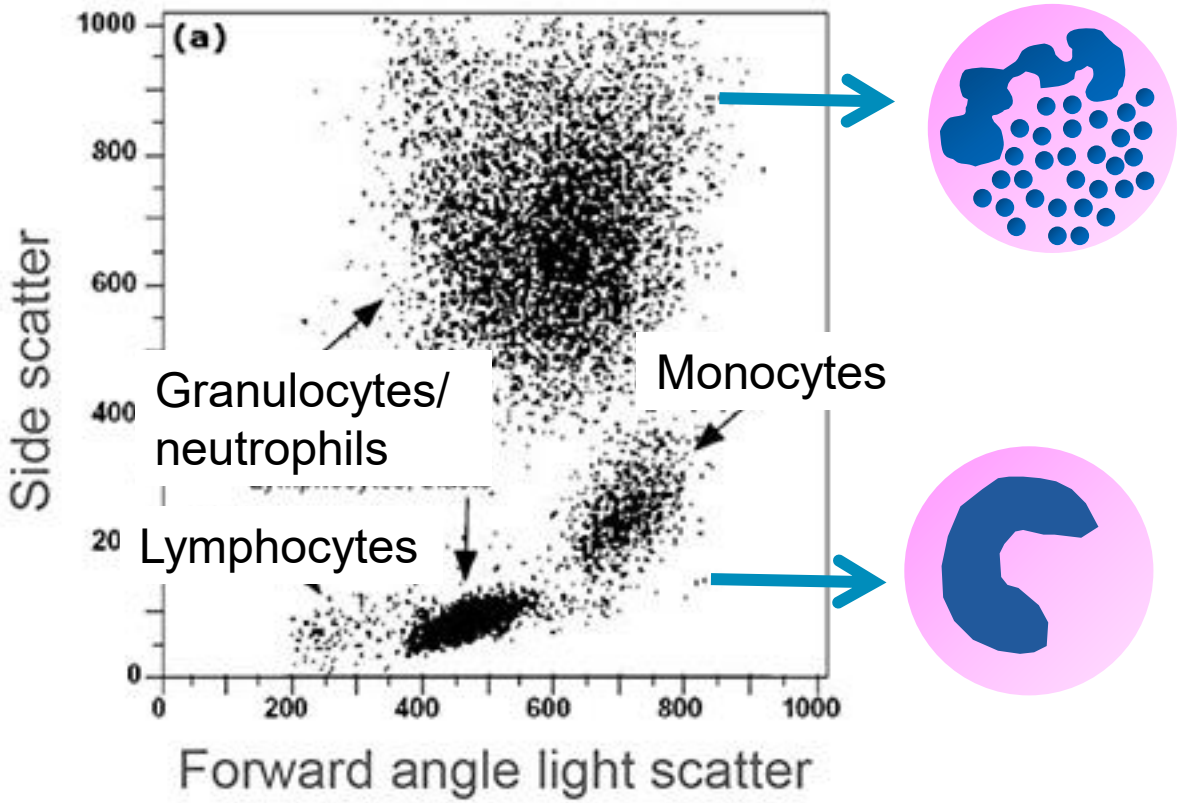
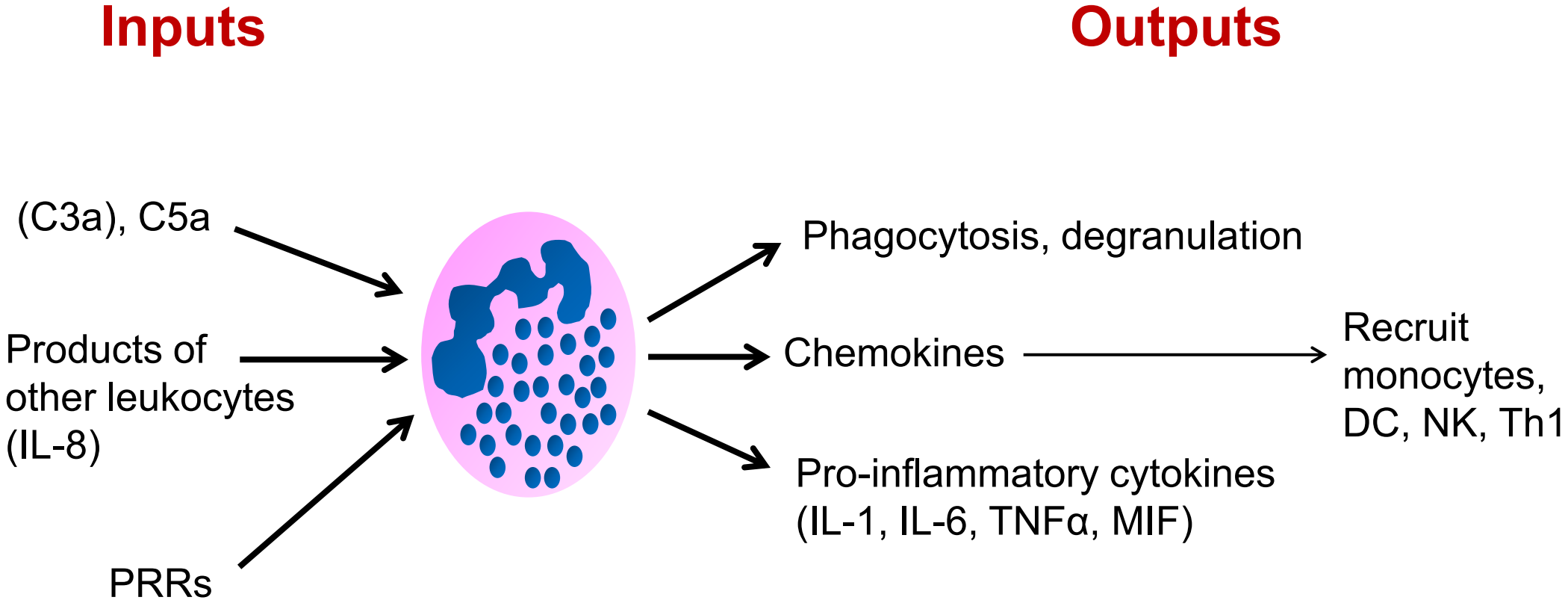
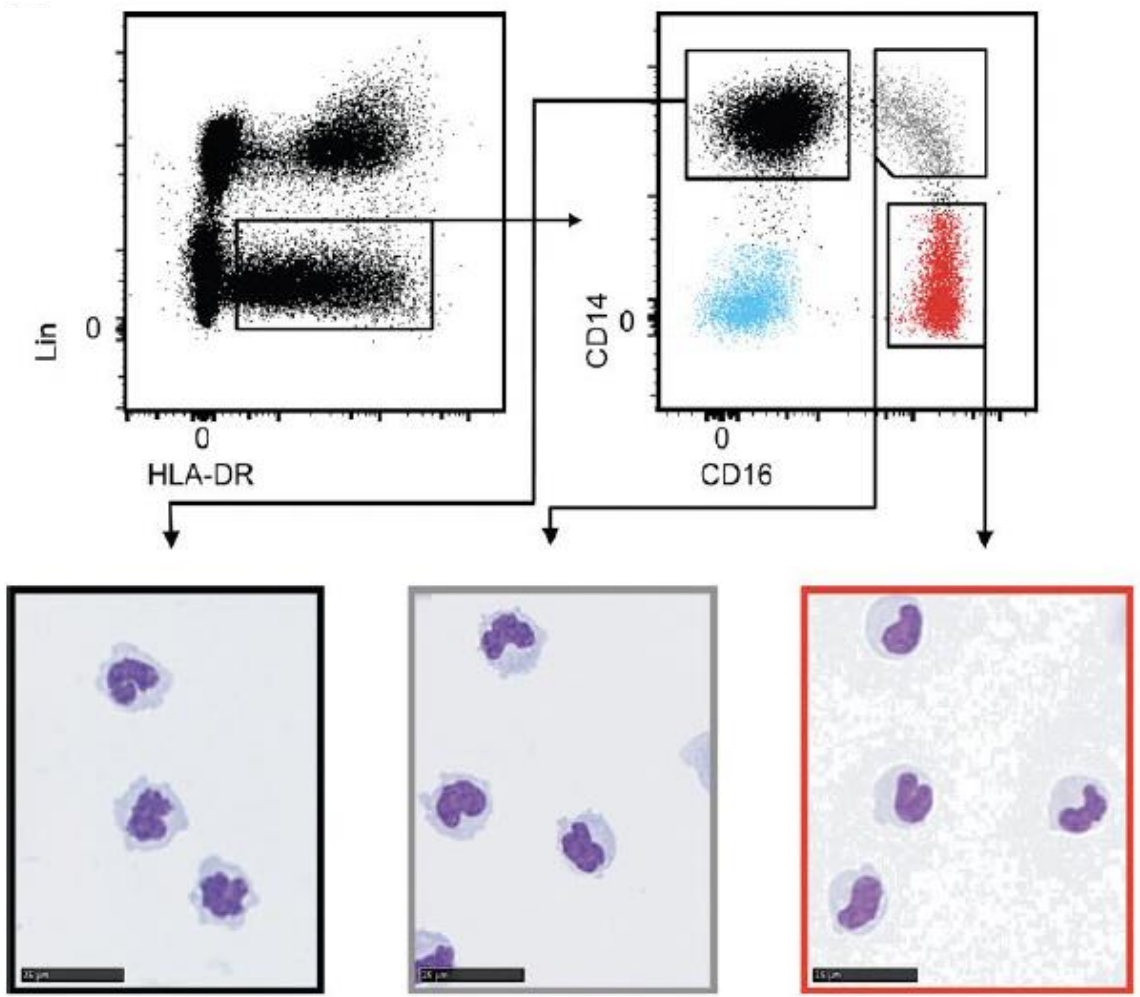


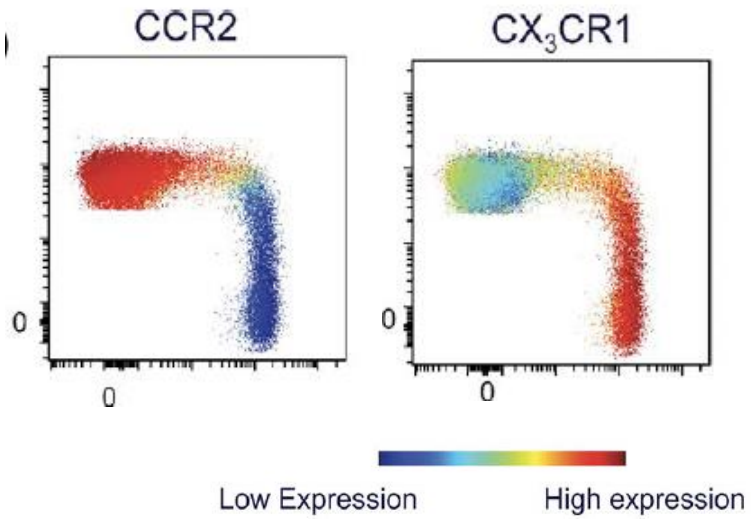
Image from AbCAM

Neutrophils (granulocytes)



Monocytes





Patel et al (2017) J Exp Med

Monocytes

Inputs

PRRs

Products of other cells (IFNα/β)

Outputs

Differentiation into monocyte-derived DC & Macs

Cytokines (TNFα, IL-6, IL-10)

Phagocytosis

Antigen presentation to T cells

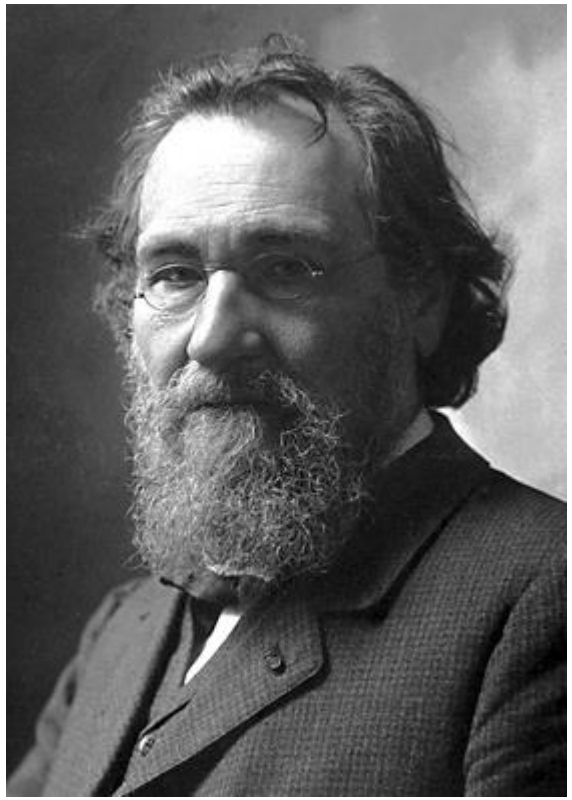
Tissue re-modelling

? Inflamed sites

Patrol blood vessels:

- Removal of debris
- Recruitment and activation of neutrophils
- Anti-viral

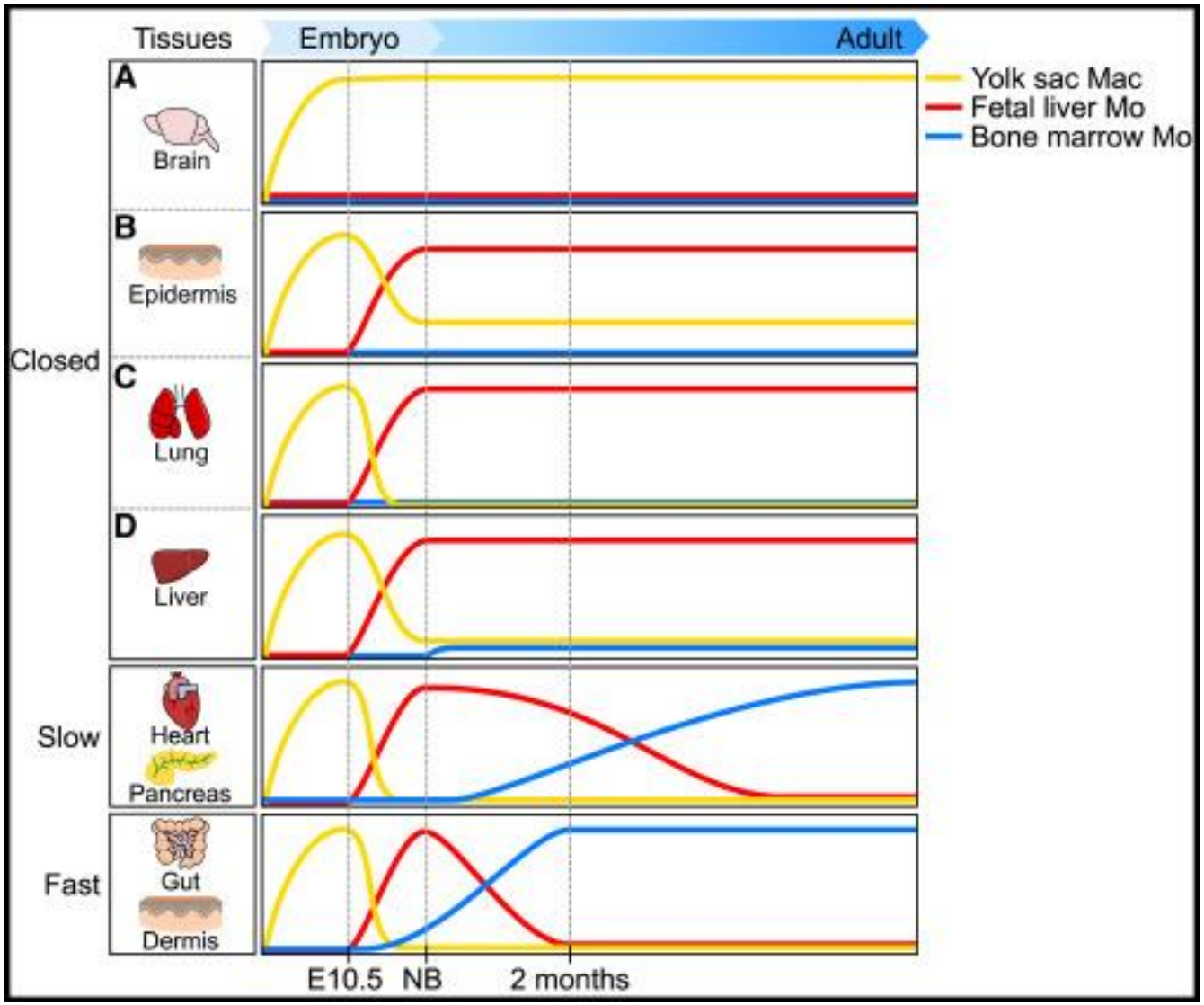
Macrophages: the big eaters



Mechnikov, 1908

Yona & Gordon (2015) Frontiers in Immunology

Tissue macrophages have different origins

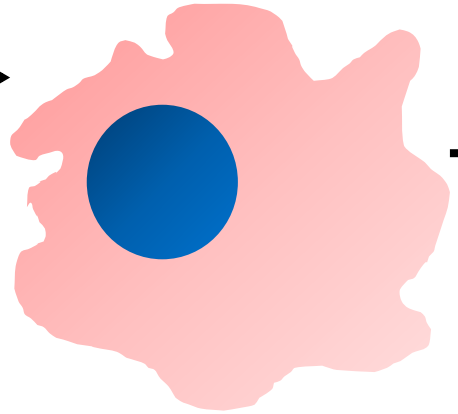


Ginhoux & Guilliams (2017) Immunity

Macrophages: phagocytosis

Inputs **Outputs**

Opsonised cells
- Antibody-coated cells →
- C3b-coated cells →

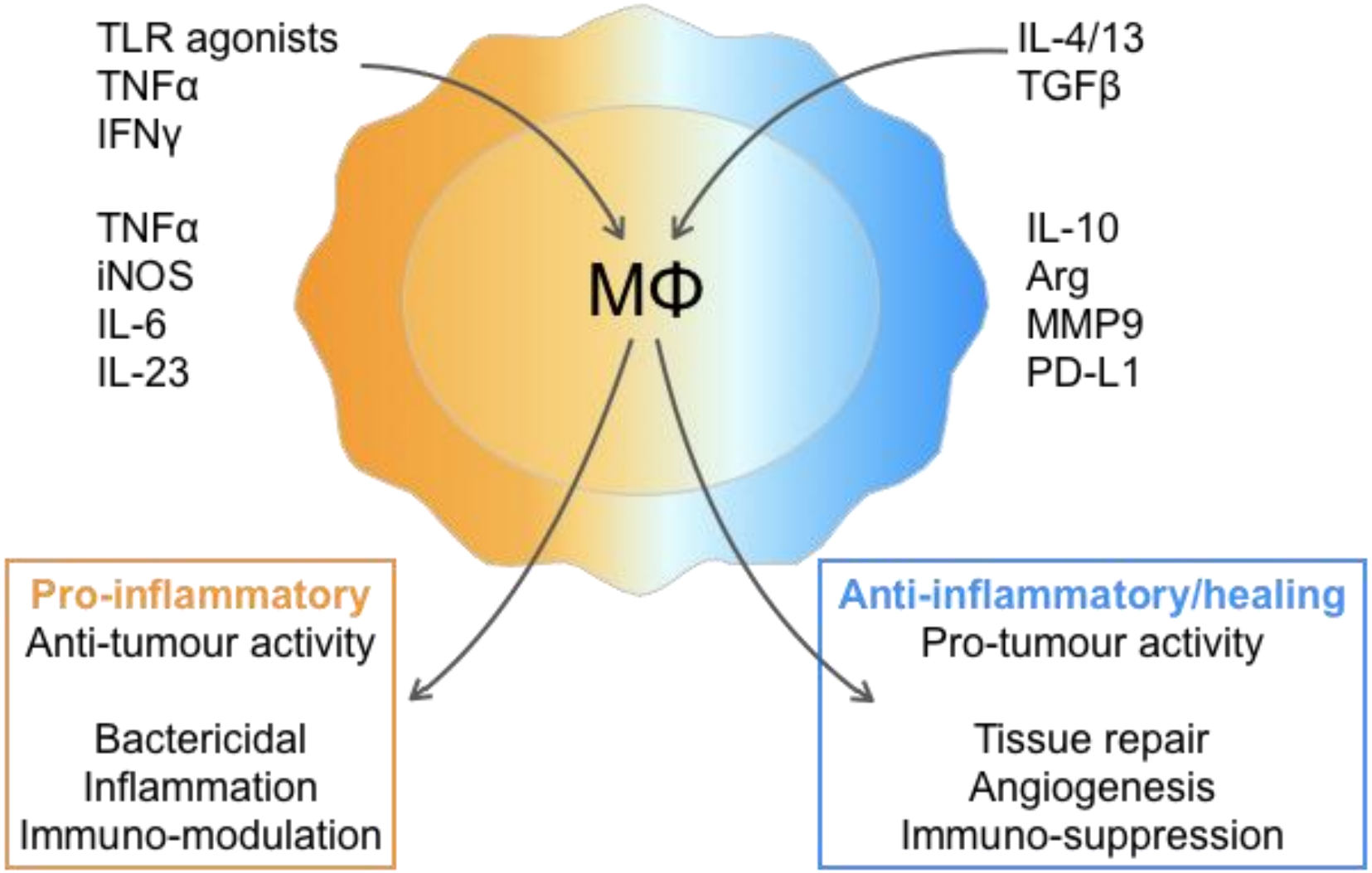


→ Phagocytosis

Kill phagocytosed cells with ROS and RNS.
IFNγ promotes this.

Can present antigen to activated T cells in the periphery.

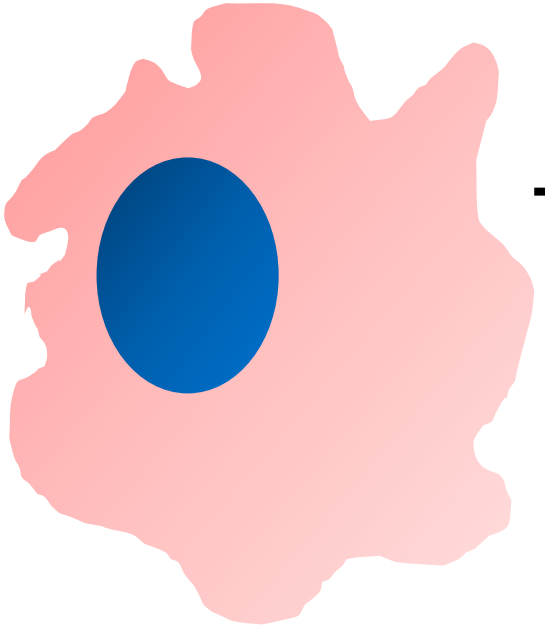
Macrophages are very responsive to the local environment



Macrophages: cytokine production

Inputs

Necrotic cells
Pathogens
(PRRs)



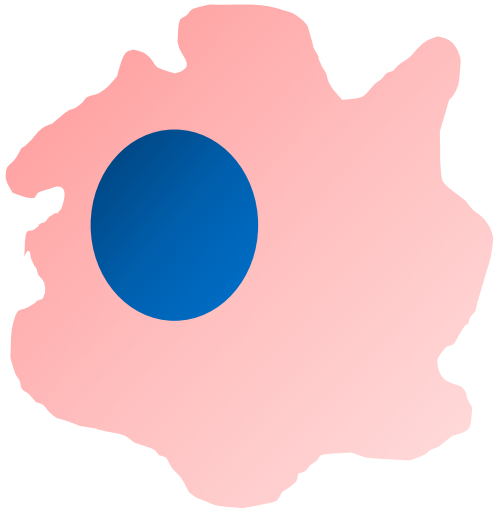
Outputs

Pro-inflammatory cytokines
(IL-1, IL-6, TNFα)

Macrophages: cytokine production

Inputs

Apoptotic cells
(phosphatidylserine) →

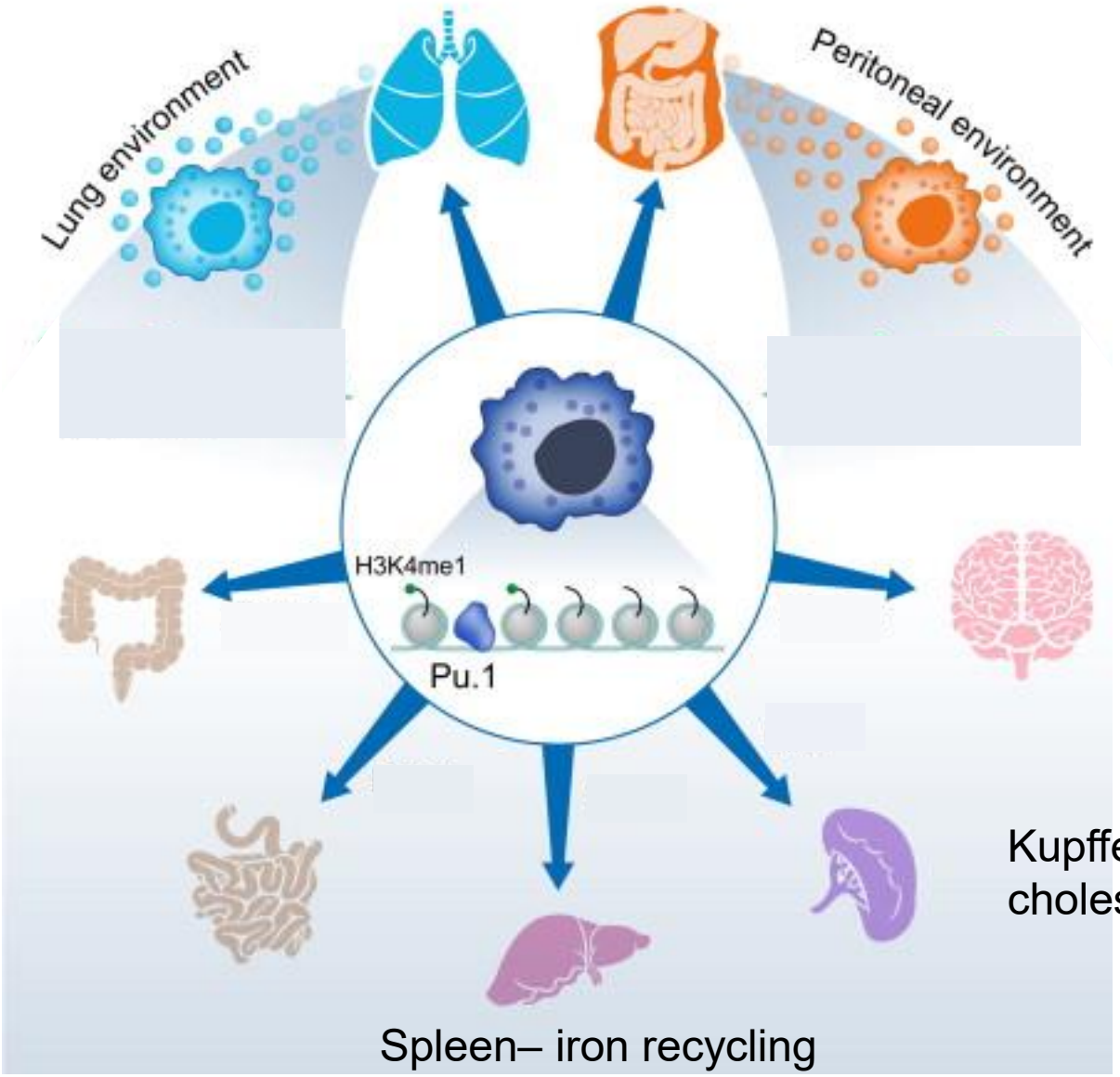


→ **Outputs**

Anti-inflammatory cytokines
(IL-10, TGFβ)

Tissue-resident macrophages have specialised functions

Alveolar macs– phagocytosis of excessive surfactant



Colon– tolerance

Microglia – synaptic pruning, clearing of debris

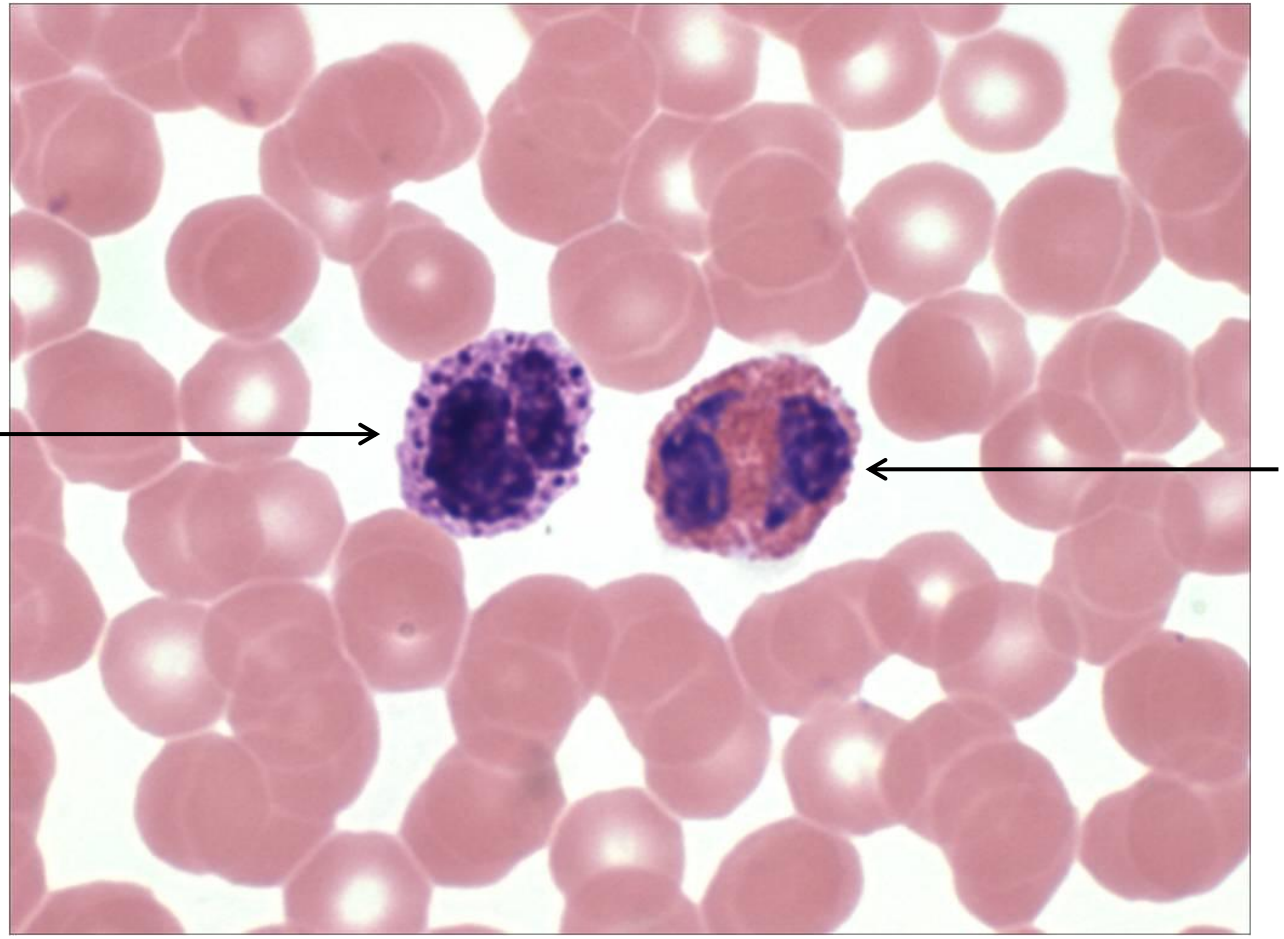
Small intestine – tolerance

Kupffer cells – iron and cholesterol recycling

Spleen– iron recycling

Lavin et al (2014) Cell

Eosinophils and basophils respond to parasite infections



Mast cells (and basophils)

Inputs

IL-3, IL-5, G-CSF →

C3a/C5a →

IgE-coated particles →

Outputs

→ Heparin
Histamine

→ IL-4

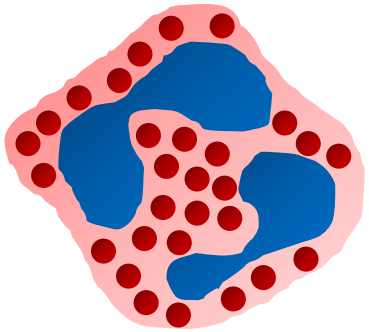
Paul Ehrlich, 1908

Eosinophils

Inputs

IL-3, IL-5, G-CSF →

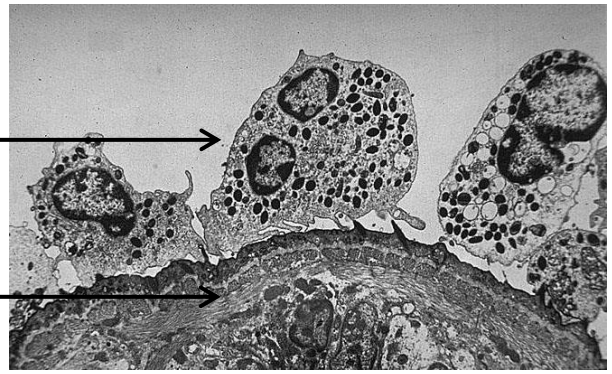
IgE-coated particles →



Outputs

→ Toxic granules (eg.MBP)

→ IL-3, IL-5, IL-8



Summary 2

1. Myeloid cells can be separated by flow cytometry based on their different size and shapes.

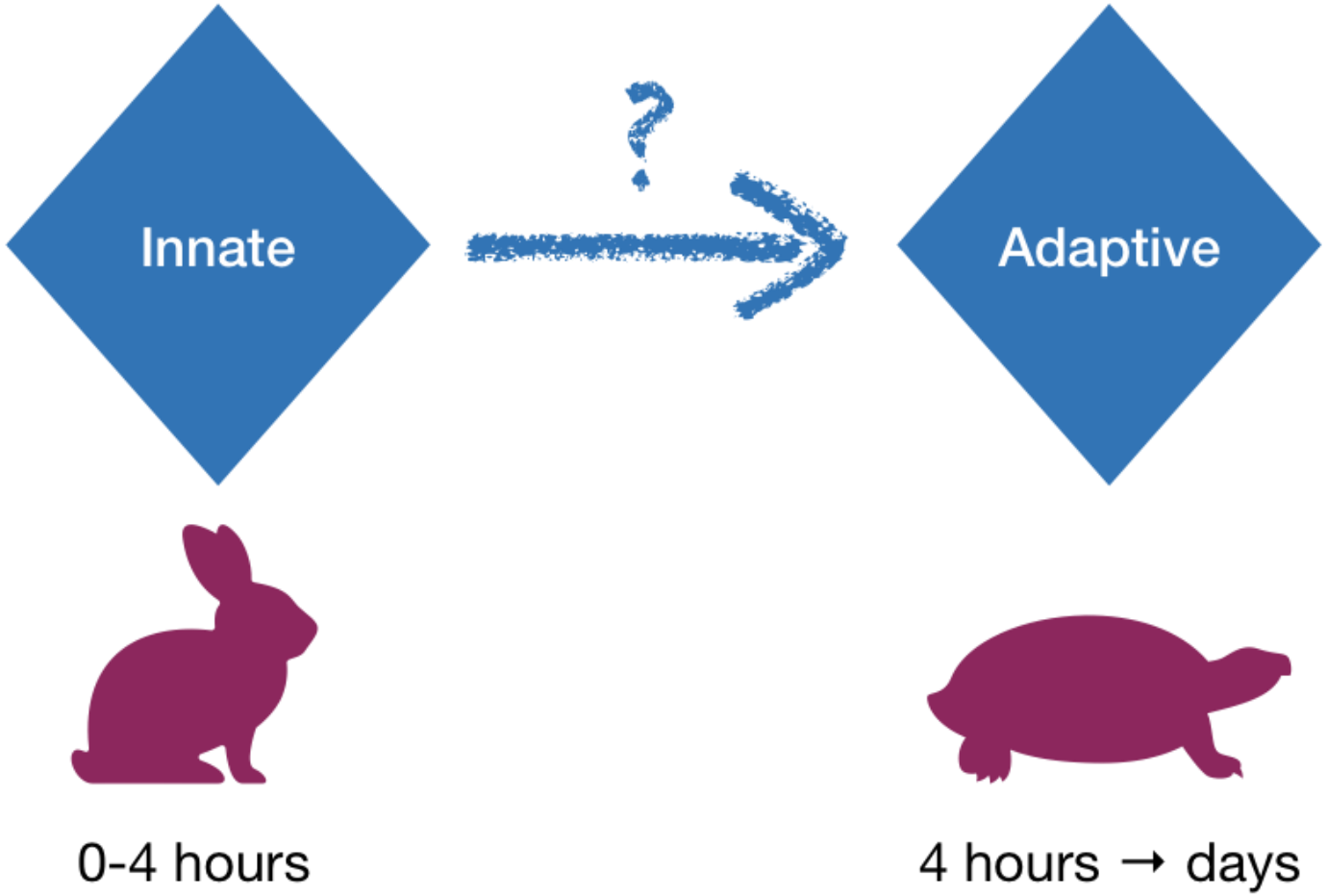
2. Neutrophils are the first cells to arrive at the scene of an infection.

3. Monocytes are effector cells, but also can become macrophages and dendritic cells.

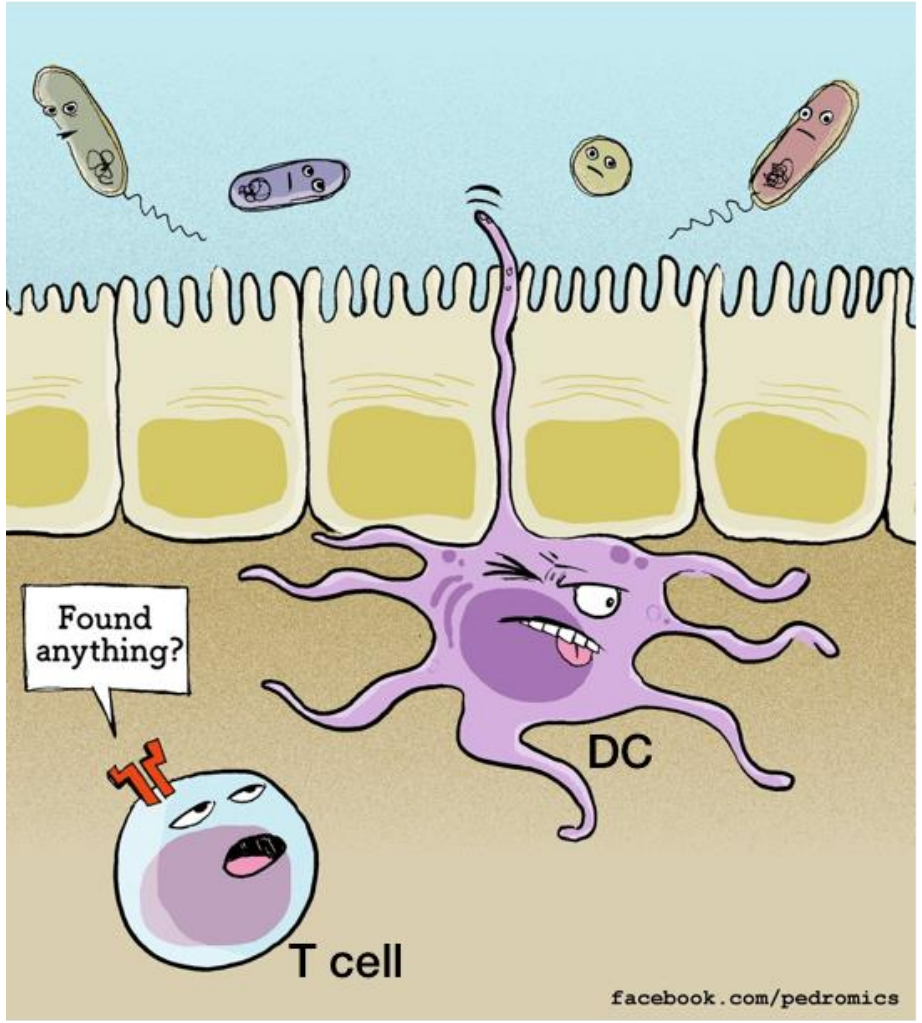
4. Macrophage function is shaped by the tissue in which they sit.

Part 3: Myeloid lineage cells

Dendritic cells

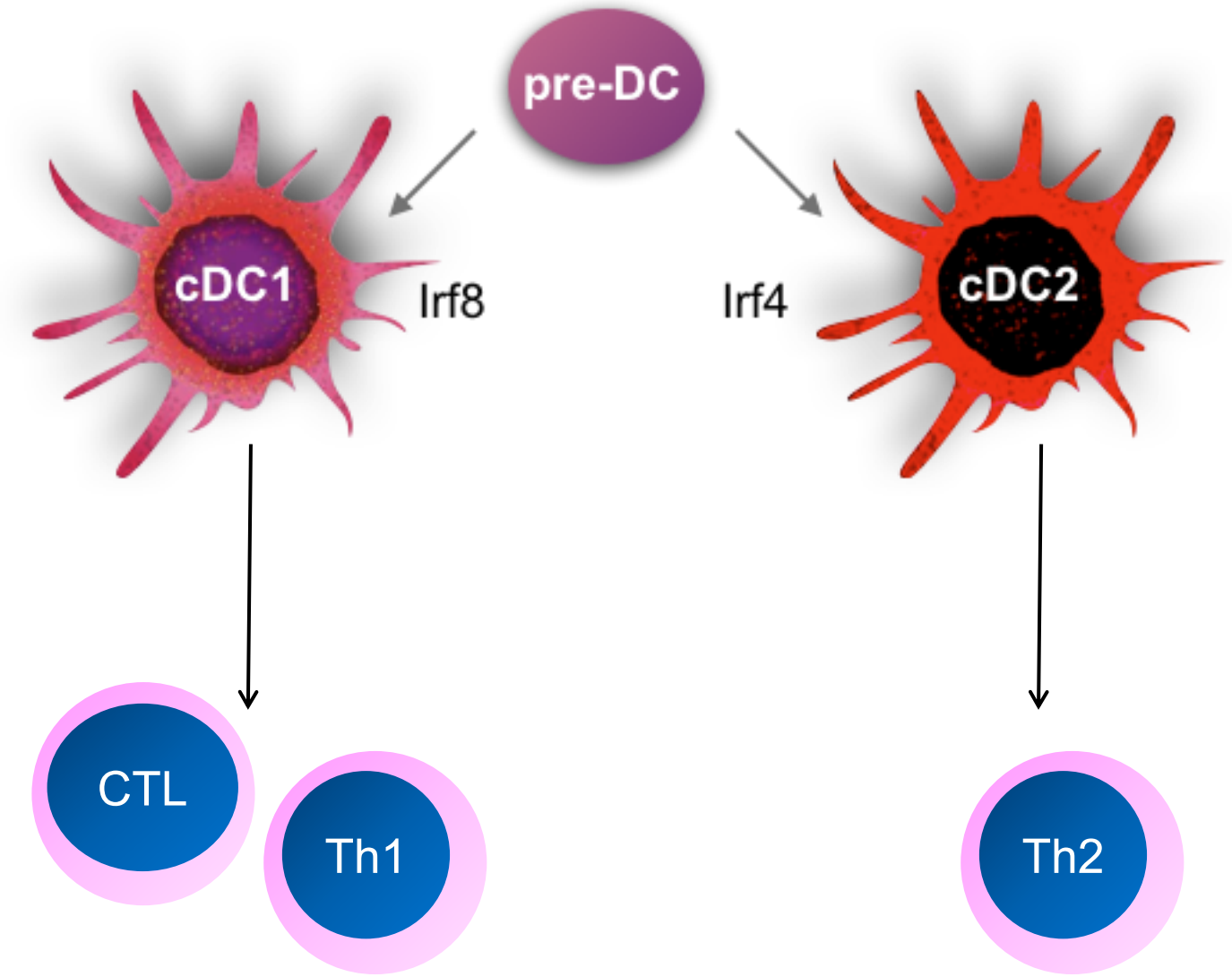

Innate
?
Adaptive
0-4 hours
4 hours → days

Dendritic cells (DC)

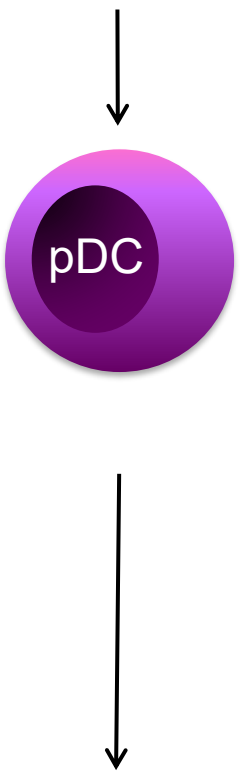


Pedro Veliça
CMB, Karolinska

There are 2 types of myeloid DC



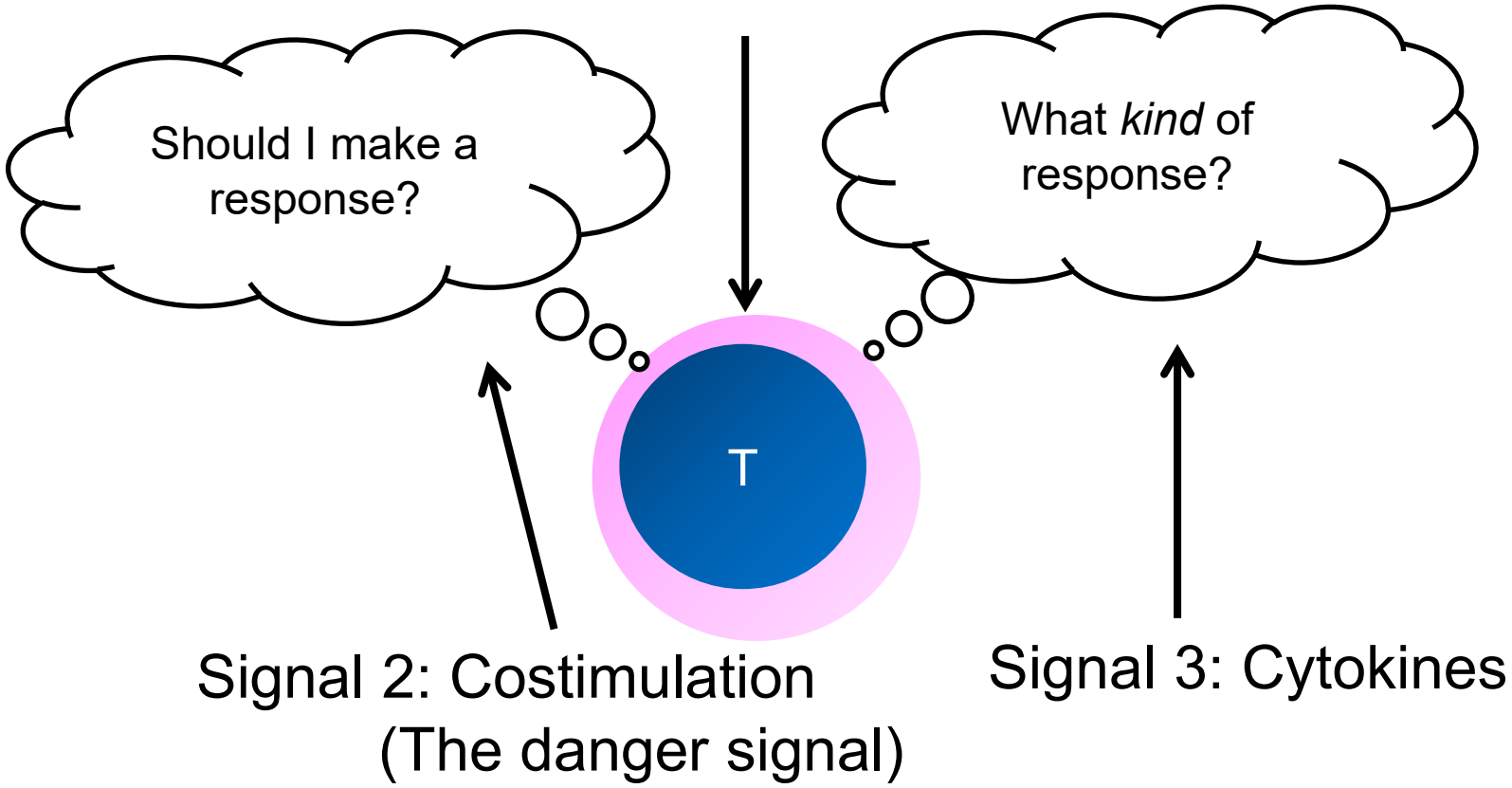
and pDC



Myeloid or Conventional DC (cDC)

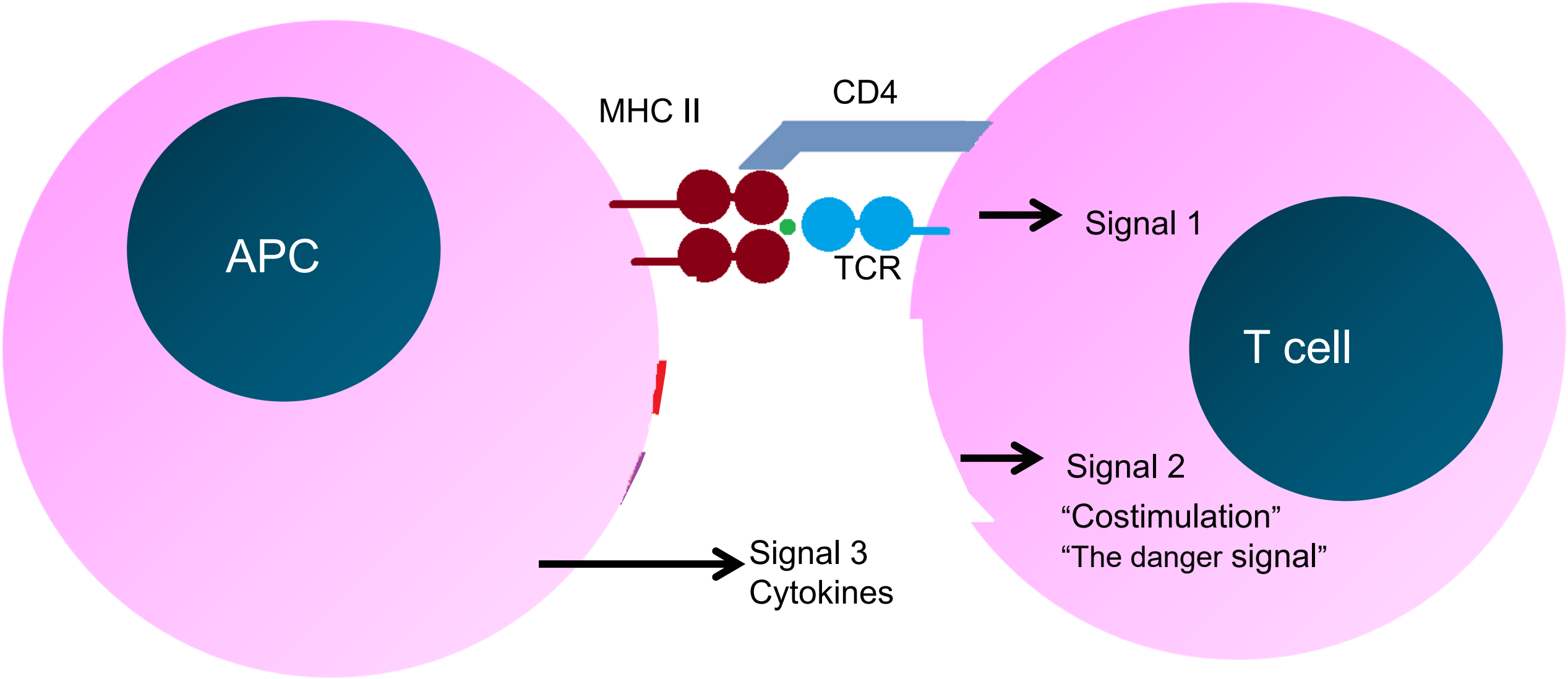
- Common DC precursor in bone marrow
- Develop into pre-DC (under the influence of FLT3L)
- Leave bone marrow as pre- committed pre-DC1 and pre-DC2 seed peripheral and lymphoid organs
- Differentiate into immature DC1 (XCR1 expressing) and DC2 (SIRPa expressing)
- Both subsets can become mature, migrate, present antigen, activate T cells

Dendritic cells help T cells to answer two questions…



Ralph Steinmann, 2011

Costimulation tells T cells whether to stop or go (signal 2: the "danger" signal)

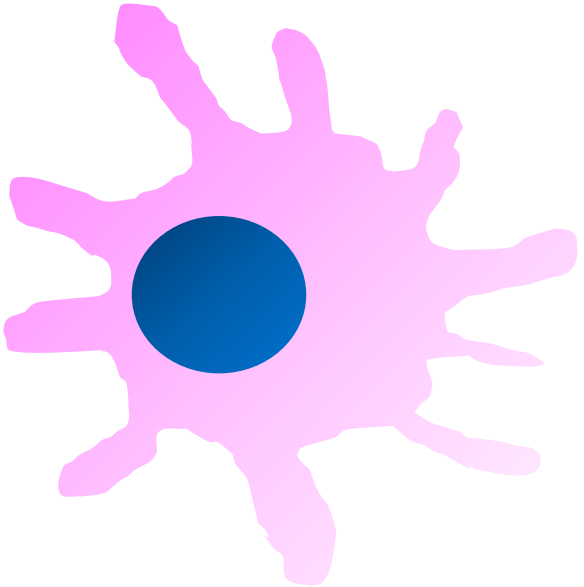


Dendritic cells

Inputs

Outputs

PRRs →



Signal 1: MHC

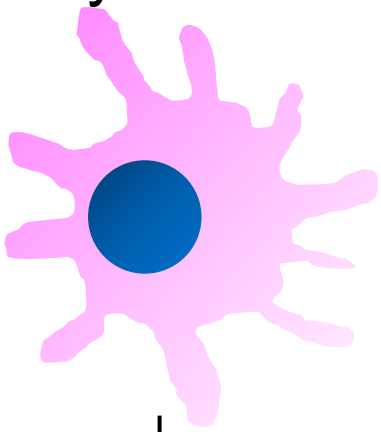
Signal 2: costimulation

Signal 3: cytokines

Dendritic cells polarise the immune response by their production of cytokines (signal 3)

Myeloid DC

TLR1, 2, 3, 5, 6, 8

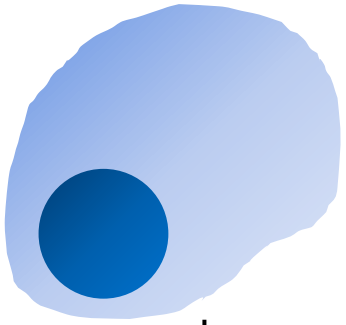


IL-12, TNFα, IL-6

IL-10

Plasmacytoid DC

TLR7, 9

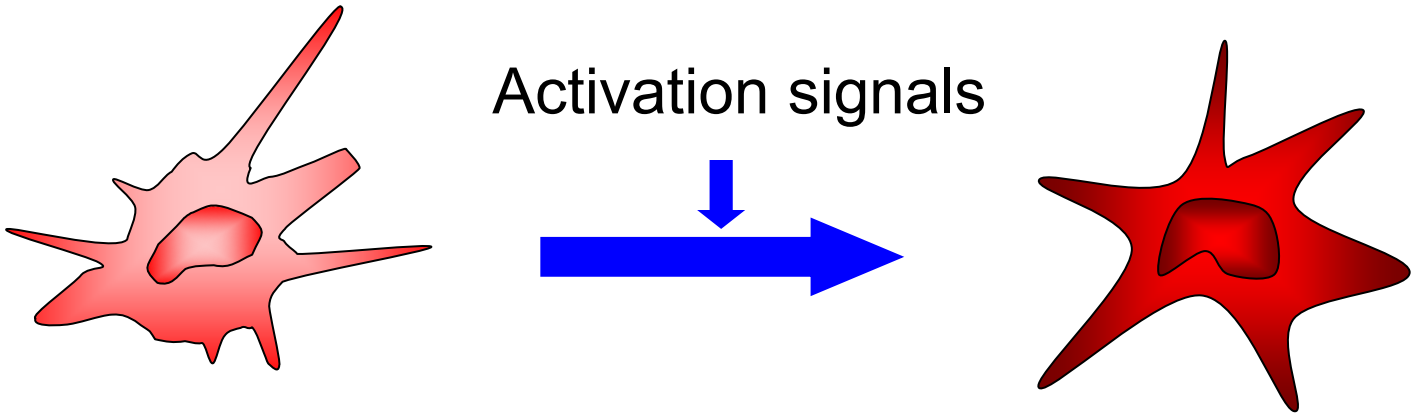


IFNα/β

Activation / maturation of Dendritic cells

Immature DC
Not "licensed"

Mature or activated DC
"Licensed" to prime a T cell response



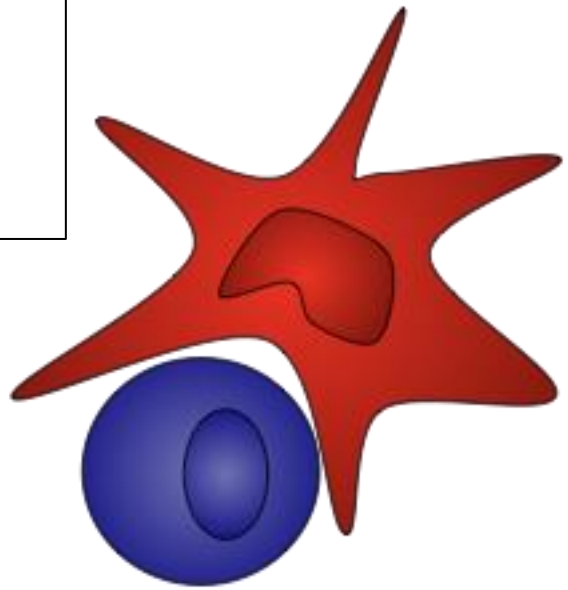
↑Phagocytosis
↑Antigen processing
↓Surface MHC I & II
↓Co-stimulatory molecules
↓Cytokines

↓ Phagocytosis
↓ Antigen processing
↑ Surface MHC I & II
↑ Co-stimulatory molecules
↑ Cytokines

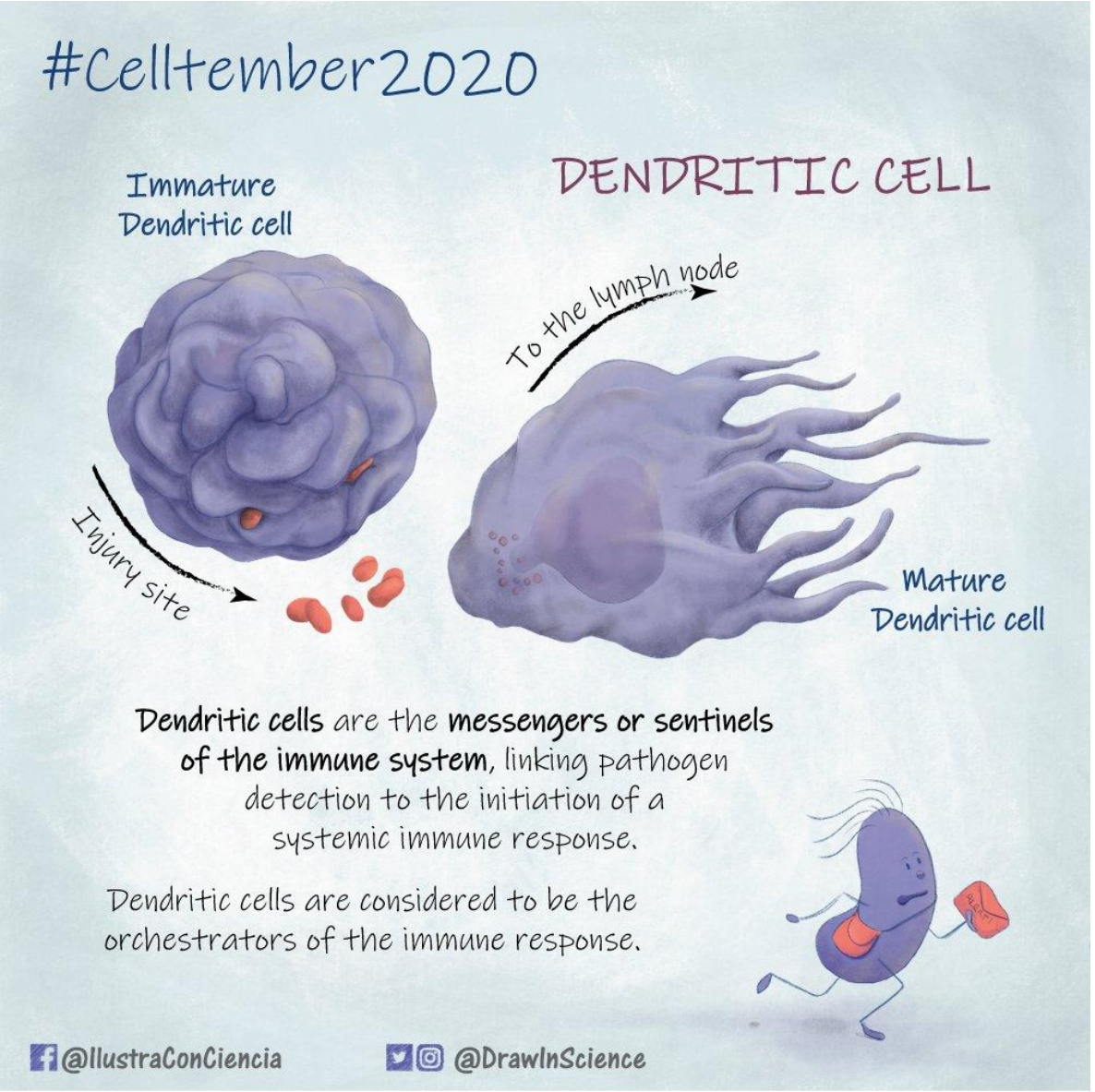
Why are Dendritic cells so good at activating adaptive immune responses?

Antigen uptake receptors and processing pathways for (cross)-presentation of MHC-peptide complexes

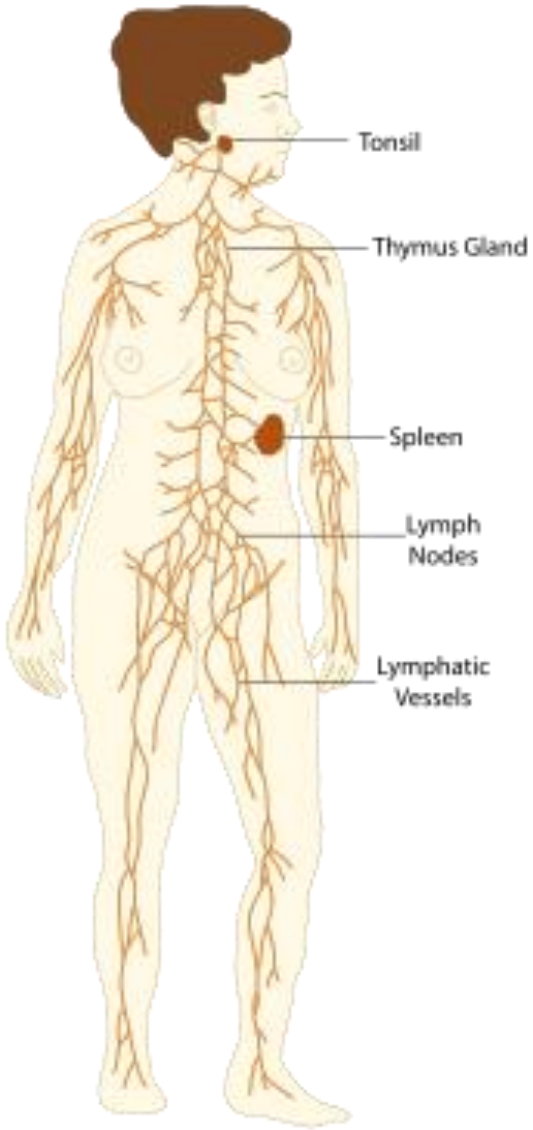
Maturation / activation in response to microbial and other stimuli



Location at body surfaces and migration to T cell areas of lymphoid organs

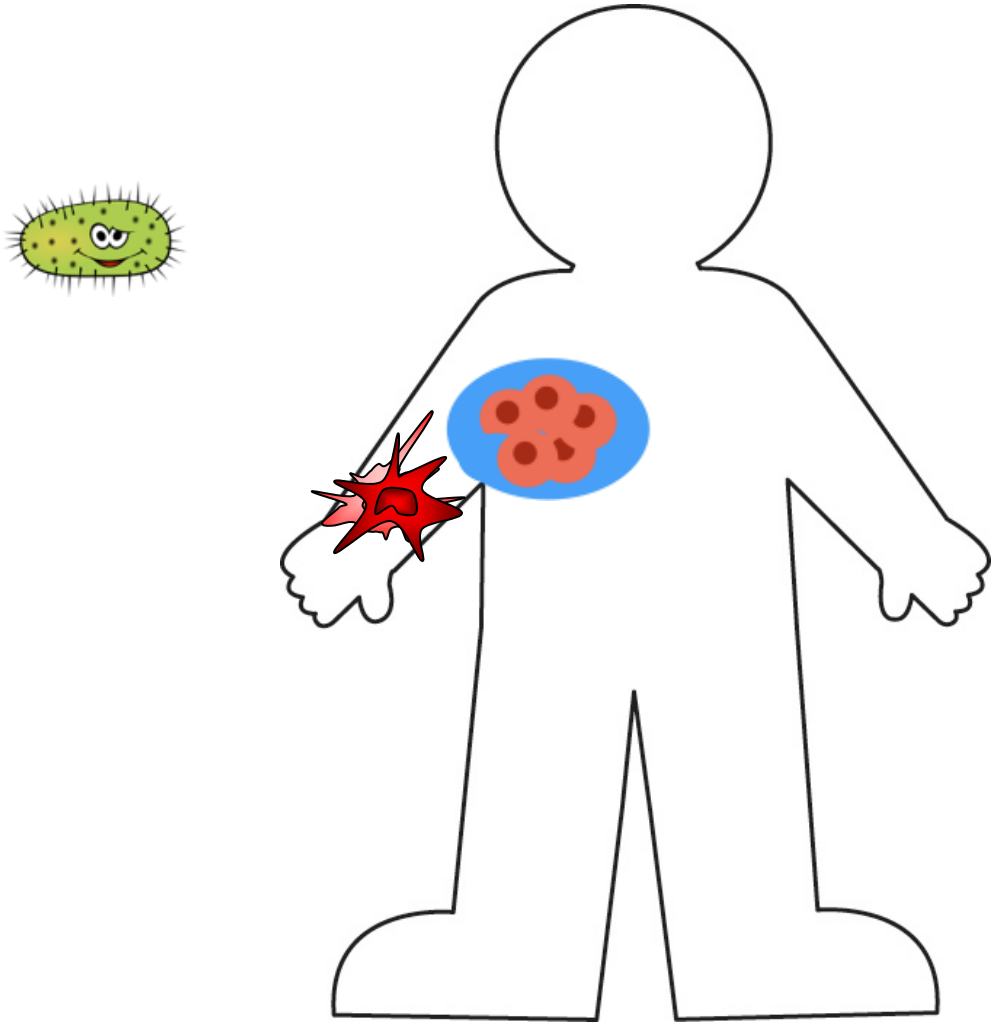

#Celltember2020
DENDRITIC CELL
Immature
Dendritic cell
To the lymph node
Injury site
Mature
Dendritic cell
Dendritic cells are the messengers or sentinels of the immune system, linking pathogen detection to the initiation of a systemic immune response.
Dendritic cells are considered to be the orchestrators of the immune response.
@IllustraConCiencia
@DrawInScience

Dendritic cells use our lymphatic system to migrate



https://en.wikipedia.org/wiki/Lymphatic_system

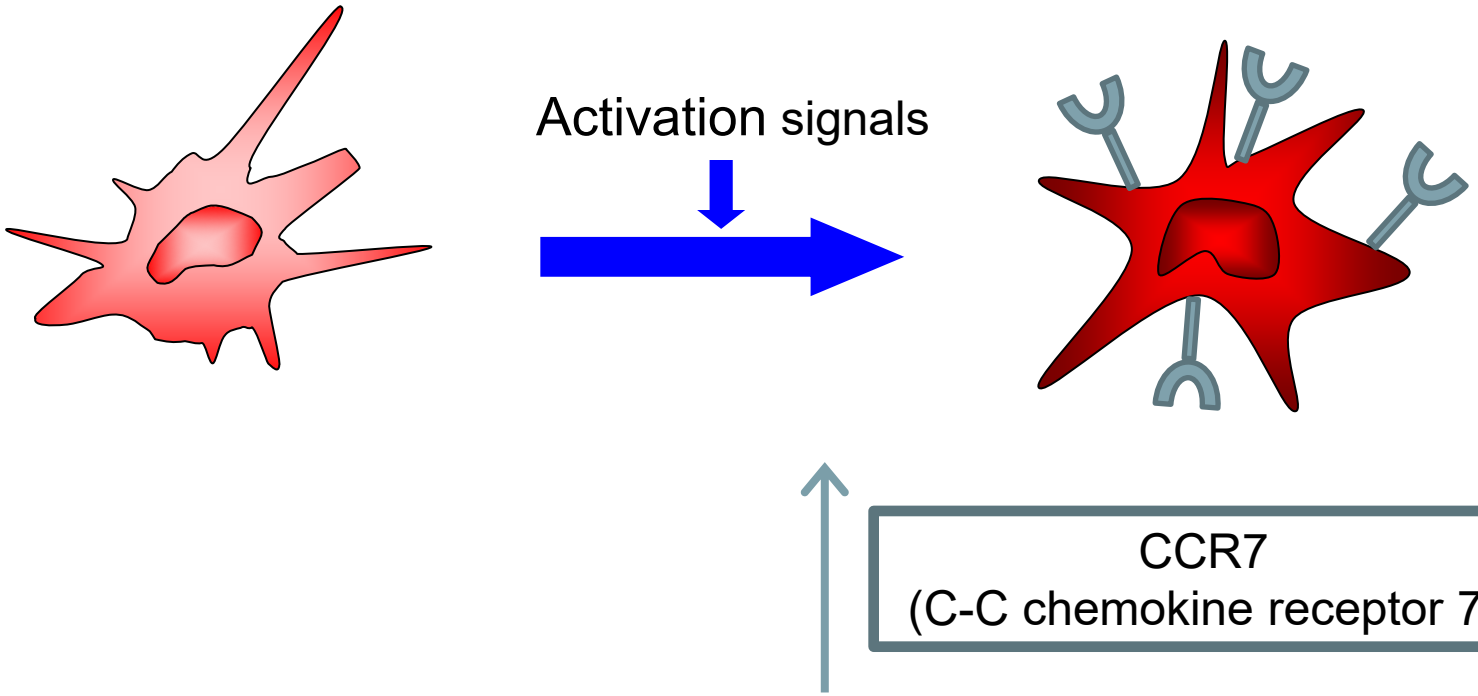
Migration of DC is linked to the presence of danger



Dendritic cells upregulate CCR7 on activation

Immature DC
Not “licensed”

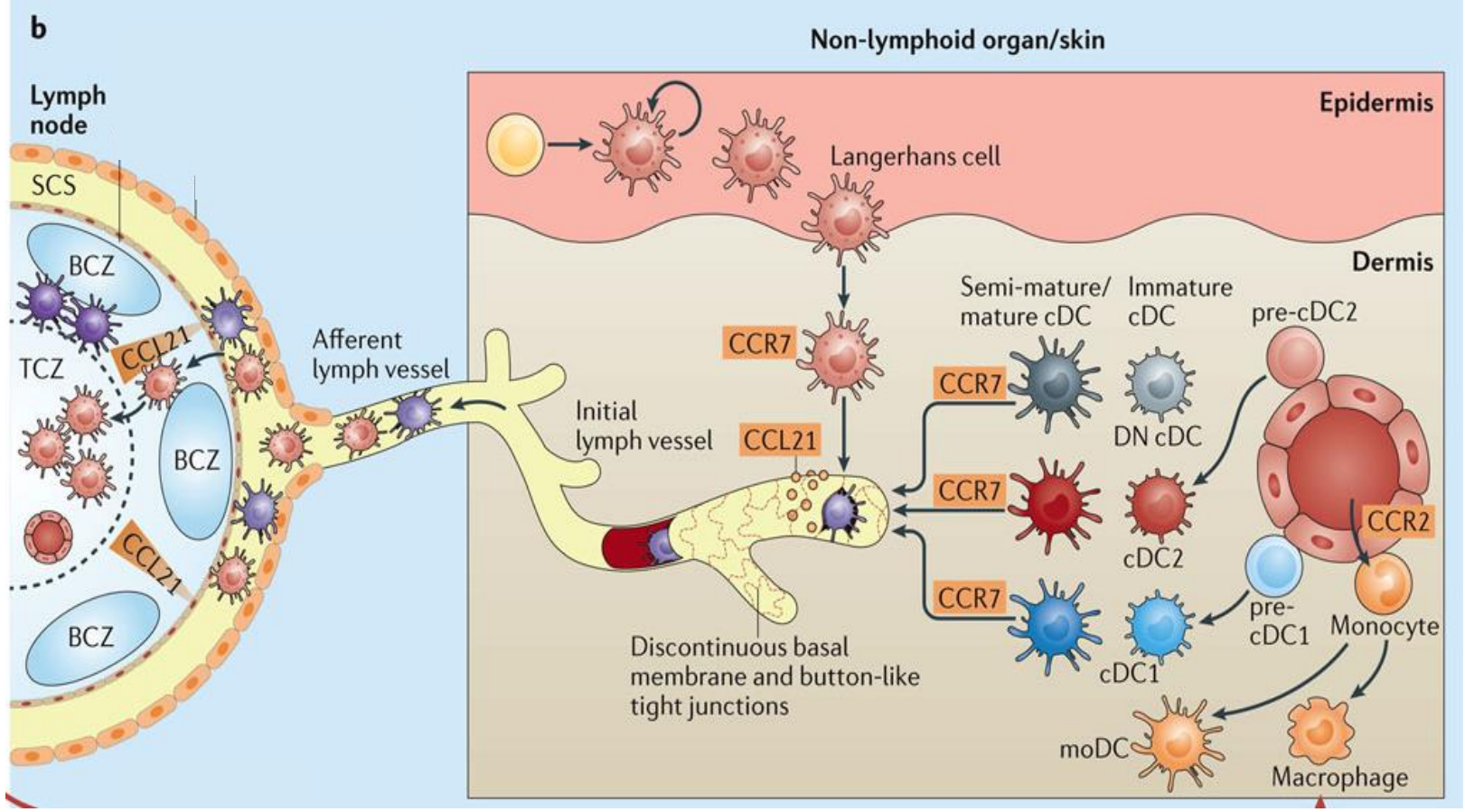
Mature or activated DC
“Licensed” to prime a T cell response



“Chemokines are chemoattractant proteins that induce cell movement”

https://www.immunology.org/public-information/bitesized-immunology/receptors-and-molecules/chemokines-introduction

Activated DC migrate to LN in response to chemokines



TCZ – T cell zone
BCZ – B cell zone

Worbs et al Nat Rev Immunol https://doi.org/10.1038/nri.2016.116

nature reviews immunology

https://doi.org/10.1038/s41577-024-01079-5

Review article

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Striking a balance: new perspectives on homeostatic dendritic cell maturation

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Summary 3

1. Dendritic cell function is defined largely by their development.

2. Dendritic cells are the bridge between the innate and adaptive immune responses.

3. Dendritic cells exist in immature and mature forms.

4. Mature, activated dendritic cells can migrate to lymph nodes and prime naïve T cells.

Function of innate immune cells: summary

	Respond to	Direct response	Onward signals
Neutrophils	C5a, IL-8, PAMPS on internalised particles	Phagocytosis Degranulation	Recruitment of other immune cells
Monocytes	PAMPs, cytokines	Phagocytosis	Pro- or anti-inflammatory cytokines
Macrophages	Opsonised particles	Phagocytosis	Pro- or anti-inflammatory cytokines (depending on internalised particle)
Basophils	C3a/C5a or IgE	Heparin and histamine release	
Eosinophils	IL-3, IL-5, IgE	Toxic granule release	IL-3, IL-5
Dendritic cells	PAMPS on internalised particles		Signal 2 (costimulation) Signal 3 (cytokines)

Useful Review Articles

Neutrophils – Kolaczkowska & Kubes (2013) Nature Reviews in Immunology 13: 159.

Monocytes - Guilliams, Mildner & Yona (2018) Immunity 49: 595.

Macrophages - Yona & Gordon (2015) Frontiers in Immunology

Basophils – Voehringer (2013) Nature Reviews Immunology 13: 362.

DC –

- Mildner, A. & Jung, S. Development and function of dendritic cell subsets. Immunity 40, 642–656 (2014).
- Cabeza-Cabrerizo, M. et al. Dendritic cells revisited. Annu. Rev. Immunol. 39, 131–166 (2021). This recent review nicely summarizes the latest insights on DC nomenclature, ontogeny and function.
- Eisenbarth, S. C. Dendritic cell subsets in T cell programming: location dictates function. Nat. Rev. Immunol. 19, 89–103 (2019).